



QUIMICA

THROUGH CHEMICAL
FIND YOUR WAY

2025
NEWS LETTER
CHEMICAL ENGINEERING SOCIETY



www.quimicabits.blogspot.com



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DIRECTOR'S MESSAGE



Dr. PANKAJ RAI,
Director, BIT Sindri

Since its founding in 1956, the Department of Chemical Engineering at BIT Sindri has stood as one of the institute's cornerstone programs and a powerhouse in Indian chemical engineering. Esteemed by industry leaders and academics alike, the department has relentlessly shaped generations of students, raising the bar for academic excellence and professional success.

Equipped with cutting-edge labs and passionate faculty, it equips students to expand their scientific thinking and master core chemical engineering concepts. Our graduates shine across sectors—landing roles at top firms, interning at elite spots like IITs and chasing international careers—while our alumni leave a lasting global impact.

Guided by the motto "Hold the Vision, Trust the Process," the department launched Quimica: a vibrant platform of blogs and articles that dive into chemical engineering's evolving world, spotlighting cutting-edge global trends. We congratulate Team Quimica and the Department on this bold step toward knowledge-sharing and innovation. Here's to Quimica inspiring future engineers for years ahead!

HOD'S MESSAGE



Dr. Amit Kumar Gupta,
Associate Professor, BIT Sindri

"Life is about uncovering your potential and unleashing it fully," as a wise voice once proclaimed—a mantra that pulses through the heart of BIT Sindri's Chemical Engineering family.

Since 1956, our department has stood as a beacon of academic brilliance, delivering cutting-edge B.Tech and M.Tech programs. Equipped with world-class labs and a commitment to well-rounded growth, we empower students to shine on the global stage.

Our graduates conquer fields like R&D, consulting and PSUs, with alumni leading at giants such as IOCL, GAIL, ONGC and Vedanta. Hands-on internships at CSIR-CIMFR, IIT-ISM Dhanbad, IOCL, Tata Steel and Reliance build real-world expertise.

We celebrate triumphs like a student's stellar AIR 50 in GATE 2024 and our prestigious NBA accreditation, ensuring program excellence.

Quimica, our signature platform, sparks creativity, collaboration and breakthroughs. Fueled by a lively student body and robust alumni ties, it delivers the pulse of chemical engineering innovation.

From the Chemical Engineering Department, heartfelt best wishes to Team Quimica for sustained success and transformative impact!

PROFESSOR IN - CHARGE MESSAGE



Dr. NIRUPAMA
Associate Professor, BIT Sindri

“QUIMICA – Nurturing Ideas Beyond Classrooms”

It gives me immense pleasure to be associated with Quimica, the annual magazine of the Department of Chemical Engineering, BIT Sindri. This platform stands as a reflection of students' enthusiasm, creativity, and intellectual curiosity beyond academics. Through thoughtful articles, technical insights, and creative expressions, Quimica captures the spirit of innovation among young minds. It serves as a medium where students confidently share ideas, opinions, and experiences. Such initiatives play a vital role in shaping well-rounded individuals prepared for future responsibilities.

In today's rapidly changing world, learning must extend beyond textbooks and classrooms. Quimica encourages students to think critically, communicate effectively, and explore interdisciplinary perspectives. It provides an opportunity to engage with emerging trends, industrial practices, and real-world challenges. These experiences help students develop leadership, teamwork, and problem-solving skills. Such exposure is essential for nurturing competent and adaptable chemical engineers.

The success of this initiative is a result of the collective efforts of students, faculty members, and the institute's leadership. The dedication and teamwork displayed in bringing out this magazine are truly commendable. I appreciate the enthusiasm and sincerity of everyone involved in this endeavor. I extend my best wishes to the Quimica team for continued growth, innovation, and excellence in the years to come.

PROFESSOR IN - CHARGE MESSAGE



Dr. ASHOK KUMAR BARANWAL
Associate Professor, BIT Sindri

“QUIMICA – Connecting Ideas, Inspiring Growth”

It is a matter of great pleasure for me to share my thoughts on the occasion of the release of Quimica's annual magazine. Such publications play a significant role in encouraging students to explore their creative abilities while strengthening their writing, analytical, and editorial skills. The magazine serves as an open platform where ideas, innovation, and academic curiosity come together in a meaningful way.

Quimica, the Association of Chemical Engineers, has consistently worked as a strong link between faculty members and students of the department. Through its various initiatives, the association has cultivated a culture of teamwork, mutual respect, and academic enthusiasm. These efforts have contributed positively to the overall growth of the department and have helped build a collaborative and progressive learning environment.

I sincerely appreciate the dedication and commitment shown by the Quimica team in bringing out this edition of the magazine. In a dynamic and ever-evolving discipline like chemical engineering, it is essential to reflect upon our learning, research, and practical experiences. This magazine provides an effective medium to share such reflections and to highlight the relevance of our field in today's world.

I encourage students to continue contributing thoughtful articles and innovative ideas in future editions. Quimica offers a valuable space for personal development, self-expression, and intellectual growth. Together, let us continue to strive for excellence and take the Department of Chemical Engineering at BIT Sindri to greater heights.



ABOUT QUIMICA

Quimica is the technical colloquium of the Chemical Engineering Department, where ideas are explored, skills are strengthened, and engineers are shaped. It is built with the vision of shaping curious minds into confident, industry-ready engineers. It brings together students, faculty, alumni, and professionals on a common platform that encourages learning beyond textbooks and inspires innovation through collaboration. With active associations with professional bodies like IChE and AIChE, Quimica stays closely connected to the evolving world of chemical engineering.

At its core, Quimica focuses on equipping students with practical skills, technical awareness, and problem-solving abilities that align with real-world industrial demands. Through thoughtfully curated workshops, webinars, blogs, and interactive discussions, the society creates meaningful learning experiences that bridge the gap between academic concepts and their practical applications.

The highlight of the year, QUIMICA'25, showcases the society's spirit of innovation and creativity. Flagship events such as Innoverse and Smart Scape provide students with an engaging arena to test their analytical thinking, present original ideas, and enjoy learning in a competitive yet fun environment.

Adding further value to this experience is Quimica's Webinar Series, where industry professionals share insights drawn from real-life challenges and emerging trends. Conducted over three days, the series offers participants a deeper understanding of the field along with career-oriented guidance that helps shape informed professional paths.

Beyond technical growth, Quimica ensures a balance experienced by the students by organising activities that offer a refreshing break from routine lectures, assignments, and projects. Backed by a supportive alumni network and continuous mentorship from dedicated faculty members, the society nurtures a culture of guidance, teamwork, and growth. Through its initiatives, Quimica continues to inspire future chemical engineers and contribute to the academic and professional excellence of the department.



OUR DEPARTMENT

The Department of Chemical Engineering at BIT Sindri, established in 1956, is one of the institute's leading departments. It is recognized as one of India's prominent centers for Chemical Engineering education, with strong recognition from industrial organizations, research institutions, and professional bodies. The department is accredited by the National Board of Accreditation (NBA).



The department offers a four-year B.Tech. program and an M.Tech. program in Chemical Plant Design. It is supported by a team of experienced faculty members who actively engage in industrial collaboration, consultancy, and applied research to enhance technical competence and practical exposure.

Hands-on learning is ensured through well-equipped laboratories, including the Unit Operations Lab, Process Control Lab, Petroleum Refinery Engineering Lab, Plastics Technology Lab, Process Engineering Lab, Environmental Engineering Lab, and the Chemical Engineering Thermodynamics Lab. The department's strength is further reflected in its skilled students and a strong global alumni network.

MISSION STATEMENT



To impart quality technical education to UG and PG courses in Chemical Engineering.



To activate and pursue research in thrust/emerging areas of technology in Chemical Engineering.



To provide consultancy services to industries and entrepreneurs.



To create qualified human resources to cater the needs of a sound national economy through developmental activities.



To make the department a center of excellence or research and development in Chemical Engineering and related fields.



Program-Educational Outcomes

PEO1: Information

To produce excellent people with basic knowledge resources in both our core areas of expertise in Chemical Engineering and the new research areas.

PEO2: Capabilities for Design:

To contribute significantly to technology by developing laboratory, design, and problem-solving skills for technical careers in addressing important issues for the country's social and economic advancement.

PEO3: Expert Concern

To make concentrated efforts to produce graduate students who are capable of finding their place in prestigious organisations and functioning as productive team members with the communication and collaboration skills as well as an appreciation for ethical behaviour required to succeed in their careers.

PEO4: Aspects of specialisation

Graduates of the program will be equipped to pursue careers in chemical engineering, related and interdisciplinary industries, and prestigious academic institutions for postsecondary education that benefits from a solid foundation in engineering or applied sciences.

PEO5: Learning on Your Own

Through education and personal growth based on their awareness of professional societies and library resources, workshops, industry visits, expert talks, industry interactions, etc., graduates will be trained to continue their professional development.



Program Specific Objectives

PSO1: Apply the principles of mathematics, science, and foundational chemical engineering knowledge to address complex challenges, solve critical issues, and design equipment for core chemical and allied industries.

PSO2: Develop new process plants by analyzing basic requirements and conducting feasibility studies. Utilize expertise in instrumentation, process dynamics and control, and process design. Leverage modern chemical engineering tools such as CHEMCAD, MATLAB, and others for modeling and simulation.

PSO3: Demonstrate readiness to work in chemical industries by identifying and addressing health, safety, legal, environmental, and cultural aspects. Excel in professional careers across chemical, petroleum, petrochemical, pharmaceutical, food, biotechnology, energy, materials processing, and other related industries and organizations.



COURSE STRUCTURE

FIRST YEAR

SEMESTER-I

S.No	SUBJECT CODE	SUBJECT	CREDITS
THEORY			
01	BSM01	Engineering Mathematics I	4
02	BSC02	Engineering Chemistry	2
03	ESEM1	Engineering Mechanics	3
04	ESED2	Engineering Drawing & Computer Graphics	1
05	BSBE1	Basics of Electrical & Electronics Engineering	3
TOTAL			13
PRACTICAL			
01	BSC02P	Engineering Chemistry	1
02	ESEM1P	Engineering Mechanics	1
03	ESED2P	Engineering Drawing & Computer Graphics	2
04	ESBE1P	Basics of Electrical & Electronics Engineering	1
05	HSM02	Communication Skills#	1
06	CCA01	Sports/NSS/NCC/YOGA/Painting/Music/Classical dance	1
07	VSMP1	Manufacturing Practices I	1
TOTAL			8
Grand Total Credits			21

SEMESTER-II

S.No	COURSE CODE	SUBJECT	CREDITS
THEORY			
01	BSM02	Engineering Mathematics II	4
02	BSP01	Engineering Physics	3
03	BSB02	Biology for Engineers	2
04	ESPP1	Programming for Problem Solving	2
05	ESME2	Materials Engineering	2
06	PCCH1	Basic Chemical Engineering	2
07	HSM02	Indian Knowledge System	2
TOTAL			17
PRACTICAL			
01	BSP01P	Engineering Physics	1
02	ESPP1P	Programming for Problem Solving	1
03	PCCH1P	Basic Chemical Engineering	1
04	CCA02	Sports/NCC/NSS/YOGA/Painting/Music/ Classical Dance	1
05	INT02	Summer Internship	2
TOTAL			6
Grand Total Credits			23



COURSE STRUCTURE

SECOND YEAR

SEMESTER-III

S.No	SUBJECT CODE	SUBJECT	CREDITS
THEORY			
01	BSC301	Transformation Techniques and Complex Variable	4
02	CHE301	Fluid Particle Operations	3
03	CHE302	Chemical Engineering Thermodynamics	3
04	CHE303	Chemical Process Calculations	3
05	CHE304	Heat Transfer	3
TOTAL			16
PRACTICAL			
01	CHE301P	Fluid Particle Operations Lab	1
02	CHE302P	Chemical Engineering Thermodynamics Lab	1
03	CHE303P	Heat Transfer Lab	1
04	CHE305P	Mini Project	1
TOTAL			4
AUDIT COURSE			
01	AUC301	Sports/NCC/NSS/YOGA/Painting/Music/ Classical Dance	-
02	AUC302P	Professional Skills	-
TOTAL			-
Grand Total Credits			20

SEMESTER-IV

S.No	SUBJECT CODE	SUBJECT	CREDITS
THEORY			
01	CHE401	Foundations of Data Science	3
02	CHE402	Chemical Engineering Fluid Mechanics	3
03	CHE403	Mass Transfer	3
04	CHE404	Process Control and Instrumentation	3
05	CHE405	Chemical Reaction Engineering	4
TOTAL			16
PRACTICAL			
01	CHE401P	Foundations of Data Science Lab	1
02	CHE403P	Mass Transfer Lab	1
03	CHE404P	Process Control and Instrumentation Lab	1
04	CHE405P	Chemical Reaction Engineering Lab	1
TOTAL			4
AUDIT COURSE			
01	AUC401	Mastering Personal Finance (Basic to Advance Strategies)	-
02	AUC402P	Sports/NCC/NSS/YOGA/Painting/Music/ Classical Dance	-
03	INT401P	Summer Internship (6-8 Weeks)	2
TOTAL			2
Grand Total Credits			22



COURSE STRUCTURE

THIRD YEAR

SEMESTER-V

S.No	SUBJECT CODE	SUBJECT	CREDITS
THEORY			
01	CHEC501	PCC I-Separation Process	4
02	CHEC502	PCC II-Chemical Technology	4
03	CHEPEI505	PE I-Renewable Energy Sources	3
04	CHEPEII501	PE II-Fertilizer Technology	3
05	COM501	Entrepreneurship Management	3
TOTAL			17
PRACTICAL			
01	COMP501	Seminar & Report Writing –I (UN Sustainable Development Goals)	1
02	COMP502	Business Communications	2
03	CHEP501	Chemical Technology Lab	1
04	CHEP502	Fluid Mechanics Lab	1
TOTAL			5
AUDIT COURSE			
01	AUC501,2,3,4	REGIONAL LANGUAGE (ANY ONE THROUGH NPTEL / SWAYAM) 1. MARATHI 2. KANNAD 3. TAMIL 4. TELUGU	-
02	AUC505	Sports/NCC/NSS/YOGA/Painting/Music/Classical Dance	-
TOTAL			-
Grand Total Credits			22



COURSE STRUCTURE

THIRD YEAR

SEMESTER-VI

S.No	SUBJECT CODE	SUBJECT	CREDITS
THEORY			
01	CHEC601	PCC I-Project Engineering and Economics	4
02	CHEC602	PCC II-Environmental Engineering and Sustainability in Process Industries	4
03	CHEPEIII608	PE III-Green Chemistry and Engineering	3
04	CHEPEIV603	PE IV-Computational Fluid Dynamics	3
05	CHEOEI610	OE I-Solar Energy	3
TOTAL			17
PRACTICAL			
01	CHEP601	Process Modelling and Simulation Lab	1
02	CHEP602	Process Equipment Design Lab	1
03	CHEP603	Environmental Lab	1
TOTAL			3
AUDIT COURSE			
01	AUC601,2,3,4,5,6,7	FOREIGN LANGUAGE (ANY ONE THROUGH NPTEL / SWAYAM) 1. GERMAN 2. JAPANESE 3. CHINESE 4. KOREAN 5. SPANISH 6. RUSSIAN 7. FRENCH	-
02	AUC608	Sports/NCC/NSS/YOGA/Painting/Music/Classical Dance	-
03	INT601	Industrial Internship (8 weeks)	2
TOTAL			2
Grand Total Credits			22



COURSE STRUCTURE

FOURTH YEAR

SEMESTER-VII

S.No	SUBJECT CODE	SUBJECT	CREDITS
THEORY			
01	CHE701	PCC I -Transport Phenomena	4
02	CHEPEV701,2, 3,4,5,6,7,8,9,10	PE-V (Any one from the given basket of PE-V)	3
03	CHEPEVII701,2, 3,4,5,6,7,8,9,10	PE-VI (Any one from the given basket of PE-VI)	3
04	CHEPEVIII701,2, 3,4,5,6,7,8,9,10	PE-VII (Any one from the given basket of PE-VII)	3
05	CHEOEII701,2, 3,4,5,6,7,8,9,10	OE-II (Any one from the given basket of OE-II)	3
06	CHEOEIII701,2, 3,4,5,6,7,8,9,10	OE-III (Any one from the given basket of OE-II)	3
TOTAL			19
PRACTICAL			
01	CHEP701	Computer Aided Design Lab	1
02	CHEP702	Petroleum Refinery Engineering Lab	1
TOTAL			2
AUDIT COURSE			
01	AUC701	(THROUGH NPTEL/SWAYAM) Exploring Human Values: Visions of Happiness and Perfect Society	-
02	AUC702	Sports/NCC/NSS/YOGA/Painting/Music/Classical Dance	-
TOTAL			-
PROJECT			
01	CHEP703	Minor Research Project	3
TOTAL			3
Grand Total Credits			24



COURSE STRUCTURE

FOURTH YEAR

SEMESTER-VIII

S.No	SUBJECT CODE	SUBJECT	CREDITS
		THEORY	
01	CHEP801	Major Project/ Research Project/ Industrial Internship	12
TOTAL			12
		Grand Total Credits	12



FEATURES

The department admits 120 B.Tech. and 25 M.Tech. students annually, ensuring a balanced academic environment.

CAD training in the 7th semester equips students with industry-relevant computational and design skills.

Experienced faculty mentor B.Tech. and M.Tech. projects, fostering research, analytical, and critical thinking abilities.

Well-equipped laboratories and modern facilities support practical learning, experimentation, and research.



LAB FACILITES

- Fluid Mechanics Lab
- Fluidization Engineering Lab
- Thermodynamics Lab
- Process Control Lab
- Petroleum Refinery Lab
- Process Engineering Lab
- Unit Operations Lab
- Characterization Lab





PCE Lab

Characterisation
Lab



Mass Transfer
Operations Lab



Fluidization Lab

Thermodynamics
Lab



Environmental Lab



FACULTY'S PROFILE



Dr. Amit Kumar Gupta
H.O.D & Associate Professor

Academic Background :-

PhD in Chemical Engineering, IIT Delhi
M-Tech in Chemical Engineering, IIT Bombay
B-Tech in Chemical Engineering, HBTI Kanpur

Research Area :-

Heat Transfer, Computational Fluid Mechanics,
Interfacial Science

Dr. Amar Kumar
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, VBU Hazaribagh
M-Tech in Chemical Engineering, BIT SINDRI
B-Tech in Chemical Engineering, BIT SINDRI

Research Area :-

Computational Fluid Mechanics,
Fluidization Engineering



Dr. Ajay Oraon
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT-ISM Dhanbad
M-Tech in Chemical Engineering, IISc Bangalore
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Fluid Mechanics, Separation Process



FACULTY'S PROFILE



Dr. Usha Kumari
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Kharagpur
M-Tech in Chemical Engineering, IIT Roorkee
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Pollution Control, Material Surface Engineering,
Biomass, Pyrolysis

Dr. Sunil Kumar Singh
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Guwahati
M-Tech in Chemical Engineering, IIT Kanpur
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Micro and Nano Fluidics, Biosensors,
Colloid and interface science



Prof. Manish Kumar
Assistant Professor

Academic Background :-

M-Tech in Chemical Engineering, IIT Roorkee
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Computational Fluid Dynamics, Heat Transfer,
Fluid Mechanics



FACULTY'S PROFILE



Dr. Diwakar Pandey
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT-BHU Varanasi
M-Tech in Chemical Engineering, IIT-BHU Varanasi
B-Tech in Chemical Engineering, B.I.E.T. Jhansi

Research Area :-

CO₂ capture, Bio-gas purification,
Industrial Pollution Control

Dr. Abhishek Anand Hembrom
Assistant Professor

Academic Background :-

PhD in Mineral Engineering, IIT-ISM Dhanbad
M-Tech in Mineral Engineering, IIT-ISM Dhanbad
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Coal & Mineral processing & its Beneficiation,
Coal Preparation, etc.



Dr. Ashok Kumar Baranwal
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Kanpur
M-Tech in Petroleum Engineering, IIT-ISM Dhanbad
B.Sc. in Chemical Engineering, BIT Sindri

Research Area :-

Rheology of Complex Fluids, Modeling of Non-Newtonian Fluids, Computational Fluid Dynamics



FACULTY'S PROFILE



Dr. Nirupama
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Roorkee
M-Tech in Chemical Engineering, IIT Roorkee
B-Tech in Chemical Engineering, Punjab Technical University

Research Area :-

Polymer composites, Risk assessment

Prof. Pitho Hansda
Assistant Professor

Academic Background :-

M-Tech in Mineral Engineering, IIT-ISM Dhanbad
B-Tech in Chemical Engineering, BIT, Sindri

Research Area :-

Mineral Beneficiation, Chemical Technology



Dr. Ch. V. Raghunath
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT-BHU Varanasi
M-Tech in Chemical Engineering, SRM university, Chennai
B-Tech in Chemical Engineering, A.N University

Research Area :-

Air pollution control, Multi component absorption, Process Development



FACULTY'S PROFILE



Dr. Devina Ratnam
Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT-ISM Dhanbad
M-Tech in Chemical Engineering, IIT Gandhinagar
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Nanotechnology synthesis of boron based nanosheets

Prof. (Ms.) Poornima Pandey
Assistant Professor

Academic Background :-

PhD (Pursuing) in Chemical Engineering, IIT-ISM Dhanbad
M-Tech in Chemical Engineering, IIT Gandhinagar
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Air pollution control, Catalytic Separation Processes





FACULTY'S MESSAGE



Prof. Manish Kumar
Assistant Professor,
Chemical Engineering Department,
B.I.T. Sindri

"Empowering Chemical Engineering's Future Through Quimica's Commitment."

The Department of Chemical Engineering is committed to fostering a culture of holistic student development, empowering graduates to compete successfully in the global professional environment. This vision is achieved through a strong academic foundation combined with continuous guidance and mentorship.

Students bring enthusiasm and fresh ideas, while teachers contribute experience and insight. It is the responsibility of educators to channel student energy in the right direction by sharing knowledge and practical wisdom. The active participation and positive response of students are truly appreciated, and sincere thanks are extended to the Quimica team for their dedicated efforts in making this initiative successful.



FACULTY'S MESSAGE



Dr. Amar Kumar
Assistant Professor,
Chemical Engineering Department,
B.I.T. Sindri

"Quimica: Shaping Chemical Engineering's Future with Knowledge and Passion."

The fact that Quimica is renewing its electronic publication for the fifth edition on behalf of the Department of Chemical Engineering at BIT Sindri, Dhanbad, is a joyful occasion. For years, Quimica has allowed our students to learn and exhibit various technical and soft skills. The Department and Quimica have made every effort to provide budding engineers with useful information and insightful direction, not just in the area of chemical engineering but also in a wider sense. Our students have made a significant influence on the national as well as international levels. In numerous national-level competitive exams like the GATE and CAT, many of our students have once again performed exceptionally well. This year, one of our chemical engineering students were awarded the esteemed MITACS scholarship and completed their three-month research internships at various renowned Canadian institutions.

FACULTY'S ARTICLE

SULPHUR REMOVAL PLANT FROM NAPHTHA



Dr Amit Kumar Gupta,
Associate Professor, BIT Sindri

Primary Sulphur Removal Plant:-

Sweet naphtha is processed naphtha with a Sulphur content of less than 50ppm and sour naphtha is processed naphtha with a Sulphur content of more than 90 ppm.

- Sour naphtha is handled in the principal Sulphur Removal Plant.
- Sweet naphtha is simply stored in overhead tanks.

This naphtha is now delivered to the vaporizer.

Vaporizer and hydrodesulfurizer (HDS) :-

- A mild steel vessel with refractory connecting serves as the vaporizer. Syngas is combined with the processed naphtha from the PSR plant before heating it from outside. Cr-Mo spiral tubes operate below 625°C, with effluent temperature crucial to the unit's performance.
- This is managed by maintaining fuel flow at a constant pressure. A basic calculation uses a temperature of 50°C, a pressure of 35 kg/cm² and a pressure drop of 2 kg/cm².
- The HDS unit is where the vaporized process naphtha goes to get its Sulphur removed. The catalyst must be manually inserted into the holding grids. It consists of a purple sandwich with two ZnO beds surrounding a CO-MO bed.

FACULTY'S ARTICLE

The rapid ZnO bed, which is created by the reaction of hydrogen and reactive Sulphur in the processed naphtha, removes the H₂S produced by the reaction between hydrogen and non-reactive Sulphur in the presence of CO-MO catalysis. When the initial ZnO bed reaches saturation, the ability to change the H₂S inlet and outflow is offered and the bed is turned around.

The HDS should not be heated above 390°C to prevent the catalyst from sintering and the generation of the unfavourable sooty carbon. On the other hand, the temperature at the inlet shouldn't be too low to prevent the need for additional fuel to heat the stream going to the main reformer.

The process naphtha from the HDS is combined with steam before going into the primary reformer. For the steam-carbon ratio, the steam flow rate is roughly five times that of naphtha. The steam carbon ratio is designed to be around 3.5.

The process naphtha from the HDS is combined with steam before going into the primary reformer. For the steam-carbon ratio, the steam flow rate is roughly five times that of naphtha. The steam carbon ratio is designed to be around 3.5.

Primary Reformer: In the primary reformer, the reforming of the process of naphtha occurs. Major reactions involved here are:



All of the above reactions are endothermic.

FACULTY'S ARTICLE

In the primary reformer, there are four rows of 33 tubes each, i.e., 132 tubes in all. The tubes are made of HKO, an alloy of Cr, Ni, and Ti additives. Pigtails are provided at the inlet and outlet of the primary reformer to release thermal stresses. The catalyst filled inside the tubes is a Ni catalyst with a ceramic carrier (alumina). No promoter is used. Fuel naphtha is burnt to generate heat in the primary reformer.

Shift Converter and Carbon Dioxide Removal:

The gas stream coming from the MGG contains CO_2 , CO , H_2 , N_2 , and other inserts. CO and CO_2 should be removed before entering into the synthesis loop due to their poisonous nature. The removal of CO is inefficient compared to the removal of CO_2 . Hence CO is first converted to CO_2 and then removed. Any traces of CO left in the process stream is consumed by hydrogen in the Methanator step.

FACULTY'S ARTICLE

COAL TAR DISTILLATION



Dr Amar Kumar,
Associate Professor, BIT Sindri

The coal tar produced by coking coal is a dark, viscous liquid with distinctively foul odours. It was once seen as a nuisance in the production of coal gas in addition to being a waste product. However, it was discovered to be a secret treasure of numerous aromatic compounds that are currently used as the raw material for lovely dyes, medications, explosives, photographic developers, goods and many other items that are used in our everyday lives.

COAL TAR DISTILLATION:

At least 200 aromatic compounds can be found in coal tar, but the following are the most significant ones:

- Benzene, toluene, xylene, naphthalene, anthracene and other hydrocarbons.
- Xylenols, cresols and other phenols.
- Quinoline and Pyridine are basic nitrogen compounds.

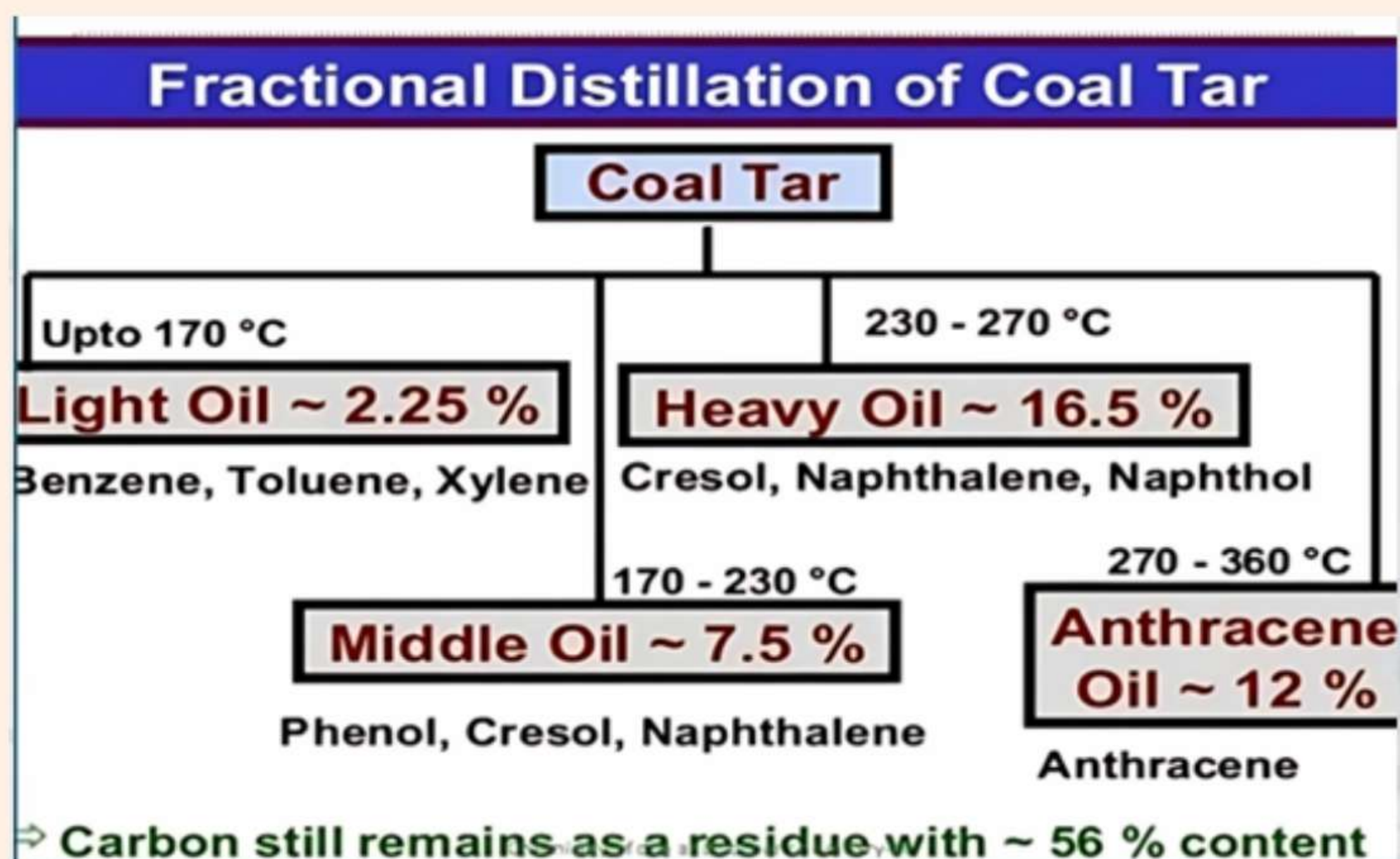
The fractional distillation process separates the components from coal tar. Following the removal, the real distillation is done in a sizable wrought-iron still with a steam pipe installed within that is heated below by fire. Connected to the still head is a condensing worm that is submerged in water.

FACULTY'S ARTICLE

Process steam is introduced into the still towards the end of the steam to prevent the residue from caking. Condensed goods are transported into various receivers from the heater. The table below lists the cuts made to the typical technical distillation of tar:

Fraction	Distillation Temp (°C)	SP.	Chief Constituents
Light Oil	Up to 170 °C	0.9	Benzene, Toluene, Xylene, Naphthalene, Anthracene
Middle Oil	170-230	1.0 to 1.02	Phenols, Naphthalene
Heavy Oil	230-270	1.02 to 1.1	Cresols, Phenols
Anthracene Oil	270-360	1.1	Anthracene

Each fraction is worked and separated into its constituents:



FACULTY'S ARTICLE

1. First Fraction (Light Oil, up to 1700C): It has benzene, toluene, and xylenes as well as some phenol and certain basic compounds, such pyridine, and is named by the fact that it is lighter than water. In order to eliminate bases and excess acid, the light oil is first washed with concentrated H_2SO_4 before being cleaned with $NaOH$. Three ingredients are recovered from the fractionally distilled, thusly purified oil:

Name of fraction	Temperature Range ($^{\circ}C$)	Chief Constituents
90% Benzol	Up to 110	Benzene, toluene, xylene
50% Benzol	110 to 140	Benzene toluene xylene
Solvent Naphtha or Benzene	140 to 170	Xylene, Mesitylene

All the 3 fractions are read under their commercial names as water of coal, solvents and for dry cleaning, 90% Benzol is fractionated again to yield Benzene ($800^{\circ}C$), Toluene ($1100^{\circ}C$) and Xylene ($1400^{\circ}C$).

2. Following Fraction (Middle Oil, $170^{\circ}C$ - $230^{\circ}C$): Its name, carbolic oil, refers to the fact that it mostly contains naphthalene and carbolic acid. Naphthalene crystals are formed when it cools, leaving behind crude liquid carbolic acid. $NaOH$ and dilute H_2SO_4 are used to purify naphthalene before it is finally sublimed. In order to dissolve the phenols, the crude carbolic acid is stirred with a $NaOH$ solution. In order to extract free

FACULTY'S ARTICLE

phenols, the solution is runoff and treated with weak H_2SO_4 , this produces crystals of carbolic acid upon distillation. It is possible to create explosives, medicines, and colours with naphthalene. Phenol is used to create colours, antiseptics, explosives, bakelite and other products.

3. Third Fraction (Heavy Oil 230°C –270°C):

It is heavier than water. Its chief ingredients are carbolic acid and cresol. It is not purified further and is used as much under the name of cresol oil, for the preservation of timbers and for making disinfectants.

4. Fourth fraction:

"Green oil", which is primarily anthracene and ranges in temperature from 270°C to 360°C. When it cools, anthracene crystals, the raw ingredient for alizarin dyes, deposit themselves. Pitch is run out of the still while it is still hot. Still residue. It is used to create stoneware that is resistant to acids, tarred paper, roofing, and road surface. It also contains 92–94% carbon.

FACULTY'S ARTICLE

CORROSION INHIBITION BY

SUPERHYDROPHOBIC SURFACES



Prof. Devina Ratnam
Associate Professor, BIT Sindri

The most widely used designed material that is essential to many applications, from industrial equipment to building construction, from marine to aeronautical uses, is metals and alloys. Metals are extremely useful because of their wealth in intrinsic qualities, including great ductility, high mechanical strength but lightweight, malleability, and a high boiling point. However, rust restricts their use and lifespan by causing disintegration. Corrosion caused premature equipment failure, product pollution, fatalities, and financial loss. Given this ongoing problem, conventional techniques have been employed to prevent metal from degradation. However, the cathodic protection [2] approach is constrained by the high cost and widespread use of non-ferrous metals. Furthermore, an innovative approach to prevent corrosion in strong corrosive settings is to coat metal surfaces with an anti-corrosion layer.

Superhydrophobic surfaces (SHS) are highly water-repellent, and their capacity to self-clean has prompted researchers to learn more about this intriguing platform for preventing corrosion [5]. Polar water makes the least amount of contact with the surface, thanks to SHS's capacity to trap air in a micro or nanotextured surface. The surface's superhydrophobicity shields it from electrochemical deterioration as well.

FACULTY'S ARTICLE

Coatings have been bio-mimetic as a means to build a noble solution to corrosion. These coatings were inspired by nature, including lotus leaves, taro leaves, water striders, etc. They display contact angle hysteresis (CAH) of 10 and water contact angle (WCA) >150 . Because they can clean themselves, these surfaces have received a lot of attention due to characteristics that prevent icing, fouling, reduce drag, and prohibit corrosion.

Bioinspired, artificial SHS can be attained by two fabrication approaches:- i) surface modification by applying the low surface energy material, and ii) surface modification by introducing the hierarchical roughness on low surface energy material. Thus, surface energy and surface topography both contribute to the wetting and dewetting aspects of coating. Thus, the superhydrophobic coating acts as a corrosion barrier for Al, Cu, Mg, and steel [6] under seawater (NaCl solution) [2], basic, and acidic conditions [7]. Strong acids and bases can be resisted mechanically by TiO_2 , SiO_2 , and rare earth metal oxides. To create a superhydrophobic nanocomposite covering, Peng et al. combined epoxy resins with low surface energy fluoropolymers and implanted polytetrafluoroethylene (PTFE) nanoparticles [8]. It is discovered that the static water contact angle (WCA) is 158° and the sliding angle (SA) is 3° . It was discovered that the produced coating was stable for 6 months of storage in a room. The tape peel test and abrasion test both verified the coating's mechanical workability. Its WCA falls to 155° after 30 cycles, whereas SA underwent a modest shift of 1. Aqua regia, a highly concentrated solution of hydrochloric acid (HCl) and nitric acid (HNO_3) in a 3:1 volume ratio, and a 1M NaOH solution have both been used in experiments to test the changed substrate's chemical susceptibility. WCA only slightly changes after

FACULTY'S ARTICLE

exposure to aqua regia and basic solution for 6 hours and 12 hours, respectively, but is still maintained at >150 . These findings demonstrate its chemical resistance even in an extremely caustic environment. Halloysite nanotubes (HNTS) have been studied by Jiang et al. to create vibrant, wear-resistant superhydrophobic coatings [9]. Its main shortcomings, including poor durability [10] and reduced abrasion resistance SHS, restrict its use. The coatings lose their superhydrophobic properties shortly after application due to the experimentally discovered sensitivity to impact and attrition. Durability is therefore absolutely necessary for mechanical application in big machines. As a result, it is impossible to totally halt this problem, but it is more practical and affordable to take preventive measures to stop the metal surface from corroding through SHS. Traditional corrosion inhibitors have been in use for a very long time. However, the development of self-cleaning superhydrophobic surfaces has opened up new research avenues, and achieving mechanical robustness and attrition resistance will not be sufficient to halt corrosion's loss and devastation.

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FACULTY'S ARTICLE

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STUDENT'S ARTICLE

Sangita Kumari

Chemical Engineering, 2021

MITACS RESEARCH SCHOLAR 2024

The history of all great work bears witness to the fact that no great achievement has ever been accomplished without the active or passive support of one's surroundings. Consequently, I can see how mentoring enhanced my project execution. I am deeply grateful to Professor Adango Miadonye Sir and Professor Mumuni Sir of Cape Breton University's Chemical Engineering Department in Canada for their invaluable guidance throughout the project's completion. Last but not least, I would like to express my heartfelt appreciation to my parents, professors, seniors, and fellow colleagues for their unconditional support. I also extend my gratitude to Mitacs, a research company in Canada, for their financial support.

The project I worked on was a study on the phase behavior and viscosity correction of petroleum mixtures.

STUDENT'S ARTICLE

Reducing the viscosity of crude oil for efficient pipeline transportation is of paramount importance in the oil industry. Initially, the viscosity of crude oil can exceed 1 million mPa-s, posing a significant challenge for recovery, production, and transportation. Various techniques, including heating, dilution, emulsification, and core annular flow, can reduce the viscosity of heavy oil.

The dilution method is widely used for transporting heavy crude oil and bitumen due to its ease of application. Organic solvents and light hydrocarbons are used as diluents to reduce the viscosity of heavy crude oil. In this research, the dilution approach was employed using toluene, Terranova crude oil, alcohols (methanol, ethanol, and propanol), Saudi oil, and their binary mixtures.

The viscosity of alcohols and toluene was measured at various temperatures (20, 25, 30, 35, and 40°C). Then, the viscosity of Terranova crude oil was measured by mixing it with different compositions of toluene at varying temperatures, resulting in a reduction in crude oil viscosity. However, the impact of diluents diminished with increasing concentrations. Toluene showed a greater impact on reducing viscosity compared to alcohols.

STUDENT'S ARTICLE

The Puttagunta et al. correlation, modified Puttagunta et al. correlation in terms of composition, and the Cragoe Equation, Lederer equation by Shu (1984) were used to predict the viscosity of mixtures of ethanol and methanol. The viscosity reduction parameter was used to compare the predicted viscosity with the measured viscosity, yielding overall values of 3.6364, 3.6364, and 18.3696.

Viscosity correction of petroleum mixtures is crucial for optimizing their flow characteristics in various industrial applications. Petroleum products, such as crude oil, bitumen, and refined fuels, often exhibit high viscosity, which can hinder transportation through pipelines and processing in refineries.

The increasing demand for oil has led to the development of large reserves of thick and heavy oil. However, transporting this oil through pipelines is challenging due to its thickness, resulting in problems such as high pressure, increased costs, clogged pipes, and lost production. Heavy oil is defined as having a certain level of thickness (API gravity of 20 or above and/or a viscosity of 100 cP). It contains more complex molecules than regular oil, which can become unstable and cause problems during transportation, leading to losses.

STUDENT'S ARTICLE

To address this challenge, researchers and engineers employ various methods to adjust viscosity. Commonly used methods for reducing the viscosity of heavy crude oil include thermal treatment, emulsification, core annular flow, and dilution with hydrocarbons and organic solvents. Thermal treatment involves heating the petroleum mixture to reduce viscosity. This method is effective but can be costly due to energy requirements and the need for specialized equipment.

Another method involves diluting the petroleum with lighter hydrocarbons or solvents. For instance, blending heavy crude with lighter oils like gas condensates or using solvents like toluene or n-hexane can significantly reduce viscosity while preserving the original properties of the petroleum.

Emulsification is another technique where the petroleum is mixed with water or surfactants to form an emulsion. Although this reduces viscosity, there are challenges associated with separating the components back into their original forms after transportation or processing. Core annular flow (CAF) is a technique used to improve the flow of oil through pipelines. Rather than directly reducing oil viscosity, CAF creates a thin layer of water along the inner pipe wall.

STUDENT'S ARTICLE

This layer acts as a lubricant, making it easier for the oil to move through the pipeline. However, maintaining consistent core annular flow can be difficult, as the flow regime often changes, and maintaining stable CAF conditions can be complex and requires careful monitoring.

Researchers are highly interested in using solvents to dilute heavy crude oil due to its practicality. This method forms the foundation of our work. A light solvent, such as toluene, can be used to reduce viscosity without changing the original characteristics of the hydrocarbon, which is an advantage over emulsification. It can also be used in different locations regardless of the climate, unlike thermal methods.



STUDENT'S ARTICLE

Alok Ranjan

Chemical Engineering, 2021

Exploring Sustainable Solutions with Maithon Power Limited and IIT(ISM), Dhanbad

I am overwhelmed with the invaluable knowledge and experience this project has brought so far. I sincerely acknowledge Maithon Power Limited and IIT(ISM), Dhanbad, for providing this incredible platform to learn and explore. This journey would not have been possible without the constant support of my family, professors, mentors, and colleagues. I am especially grateful to the entire mentorship committee for their continuous guidance and encouragement.

The project I am working on is titled "Converting Plastic Waste into Sustainable Materials Using Pyrolysis". The growing concern over the adverse environmental impact of plastic waste demands innovative solutions. This project focuses on addressing the challenge by converting plastic waste into a valuable resource.

The process involves the pyrolysis of plastic waste, which breaks it down into hydrocarbons. These residues are then combined with carefully selected components such as natural fibers, biopolymers, and recycled materials to create a sustainable composite.

STUDENT'S ARTICLE

This material exhibits enhanced mechanical, thermal, and chemical properties, making it suitable for applications in construction, packaging, and various industries. Preliminary studies have shown promising results, with the composite offering improved durability and strength while adhering to sustainability goals. Additionally, the process emphasizes energy efficiency and employs a closed-loop waste management system to minimize the environmental footprint. This mentorship program has provided me with a collaborative learning environment and the opportunity to interact with researchers, industry experts, and stakeholders. Through technical sessions, workshops, and practical insights, I have gained a deeper understanding of sustainable material development and its potential to contribute to a circular economy.

STUDENT'S BLOG



INDUSTRIAL SAFETY : A KEY TO WORK PLACE PROTECTION

By :- Shashwat Yash

Batch 2024

Department. of Chemical Engineering, B.I.T Sindri.

Industrial safety practices is a very crucial part that protect the workers from accidental injuries or illness of any sort and also it is very essential for the operations to work smoothly and productively boosting employee morale and job satisfaction.

One of the most significant advancement in industrial sector has been the technological integration in safety practices that include AI powered systems that monitor hazards in real time managing the risks involved.

Some of the common hazards for which these kind of safety practices are necessary are -

1. Hazards due to electricity which include electrocution due to exposed wiring and shocks
2. Hazards due to chemical exposure such as cleaning agents or strong acids in the industry
3. Unwanted hazards which include falls due to uneven and slippery surfaces and also the unavailability of fall protection measures etc.

Now there are some of the strategies and safety protocols for ensuring the safety of workers and so -

1. Conducting regular protocols for identifying hazards.
2. Inspection of industrial equipments and facilities.
3. Educating about the safety measures with proper training and encouraging employee participation.

Concluding this , safety is everyone's responsibility so for a flourishing business continuation these safety practices are a very essential part and by implementing these measures ,a healthy work environment can be ensured. Keeping these aspects in mind ,the future of industrial safety is very bright and through a blend of technological innovation and maintaining regulatory standards ,we can build a safer and secure environment which would in turn benefit everyone involved.

BREAKTHROUGHS IN BIOMEDICAL DEVICES OR NANOTECHNOLOGY

By :- Ayush Kumar Gupta

Batch 2024

Department. of Chemical Engineering, B.I.T Sindri.

Recent advancements in biomedical devices and nanotechnology have revolutionized the healthcare industry. These breakthroughs have transformed the way diseases are diagnosed, treated, and managed.

Biomedical Devices

One of the most significant breakthroughs is the advent of wearable biomedical devices. From smartwatches monitoring heart rates and oxygen levels to continuous glucose monitors for diabetes management, these devices provide real time health data, enabling early detection and intervention. More advanced systems, such as implantable biosensors, are being developed to monitor chronic conditions and even deliver medication automatically, ensuring consistent and effective treatment.

Nanotechnology

Nanoparticles have been engineered to target cancer cells, reducing the harm to healthy cells. This approach has improved the efficacy of chemotherapy and radiation therapy. Researchers have also developed nanoparticles that can detect cancer biomarkers, enabling early diagnosis and treatment. Additionally, advancements in nanotechnology are driving the development of lab-on-a-chip devices. These tiny platforms can perform multiple laboratory tests on a single microchip, enabling rapid diagnostics for diseases such as HIV, tuberculosis, and even COVID-19. This is particularly valuable in resource-limited settings where traditional lab infrastructure may be unavailable.

STUDENT'S BLOG



BREAKTHROUGHS IN BIOMEDICAL DEVICES OR NANOTECHNOLOGY

Future Aspects

The integration of artificial intelligence (AI) and machine learning (ML) with biomedical devices and nanotechnology holds tremendous promise. AI-powered devices can analyze vast amounts of data, enabling personalized medicine and improving health outcomes. As research continues to advance, we can expect even more innovative breakthroughs in biomedical devices and nanotechnology.

These breakthroughs highlight the transformative power of biomedical devices and nanotechnology, promising a future where healthcare is smarter, more accessible, and profoundly impactful.

INDUSTRIAL AND SAFETY PRACTICES

By :- Nikhil James Lakra

Batch 2024

Department. of Chemical Engineering, B.I.T Sindri.

Industrial safety practices is a very crucial part that protect the workers from Industrial and safety practices refer to some guidelines, protocols, and measures put in the industries and workplaces ensuring that the employees' safety, health, and well-being is not compromised, plus protecting equipment and the environment. These practices are thus very crucial in preventing accidents, injuries, and damage to properties; however, they vary according to the kind of industry.

Main elements of industrial and safety practices include :

1.Risk Assessment: The hazards that may be involved in certain activities or equipment can be identified, and the level of risk can be determined to allow preventive measures to be planned.

2.Safety Training: Workers should be trained periodically on how to handle equipment, chemicals, and emergency situations safely. This includes procedures for fire safety, first aid, and hazardous materials.

3.Emergency Procedures: Clear protocols for emergencies, such as fire drills, evacuation routes, and procedures for medical emergencies, should be in place and well-communicated to all employees.

INDUSTRIAL AND SAFETY PRACTICES

4.Maintenance and Inspection:

The equipment and machinery are checked periodically to ensure they are in good working order. This ensures that any malfunctioning equipment is detected early enough, hence posing no hazard.

5.Ergonomics: Design of workstations and tasks to reduce physical strain on the worker and prevent musculoskeletal injuries.

6.Environmental Protection: Reducing the environmental impact of industrial activities, such as proper waste disposal, reducing emissions, and conserving energy.

7.Workplace Signage and Labels: Appropriate signs and labels are to caution against potential hazards, to demonstrate safe working practices, and to provide necessary instructions.

8.Reporting and Investigating Accidents: employees are encouraged to report unsafe conditions or incidents, while accidents should be thoroughly investigated to prevent recurrence.

Thus, the implementation of these industrial and safety practices of a workplace leaves it maintaining its safety with productivity while exposing it to lesser risks due to adherence to regulatory standards.

OUR ACHIEVERS



Anjali Sah and Rahul Kumar got placements in the Mahartna Public Sector Undertaking - Indian Oil Corporation Limited.



Binit Kumar secured an All India Rank 50 in GATE-2024 and is now placed at the Maharatna Public Sector Undertaking - Indian Oil Corporation Limited.



Rahul Deo Pal secured an All India Rank 72 in GATE-2025 and is now placed as a Scientific Officer at Nuclear Power Corporation Of India Limited.

OUR ACHIEVERS



Nitish Kumar got placement in the Maharatna Public Sector Undertaking – Gas Authority Of India Limited.



Rajiv Mishra was shortlisted for BARC/TS.



Nishant Kumar secured an exceptional 99.56 percentile in the prestigious Common Admission Test examination.



TECHNIP ENERGIES INTERNS



SAMRIDDHI SINGH
(2022-26)



SANJANA SHARMA
(2022-26)



TATA STEEL INSPIRE INTERNS



**ASIF GULMOZZAM
FAYAZI
(2023-27)**



NEETU DUTTA
(2023-27)

TATA STEEL WOMEN OF METTLE



NIDHISHREE MAHATO
(2022-26)



ANUPRIYA KUMARI
(2023-27)

OUR ACHIEVERS



Reliance RELIANCE INTERNS



ABHAY KUMAR GUPTA
(2022-26)



KHUSHI BARNWAL
(2022-26)



ADITYA KUMAR
(2022-26)



KHUSHI KUMARI
(2022-26)



AKASH NATH
(2022-26)



RAVI SHANKAR PRASAD
(2022-26)



DIKSHA
(2022-26)



RUDRANIL GANGULY
(2022-26)



HARSH GIRI
(2022-26)



SAHITYA KUMAR
(2022-26)



HARSHITA SETH
(2022-26)



SAUMYA KUMARI
(2022-26)

GATE RANKERS 2025



RAHUL DEO PAL
(2020-24)
AIR 72



ABHAY ANAND
(2021-25)
AIR 1402



RAHUL KUMAR
(2019-23)
AIR 333



ASHISH MISHRA
(2021-25)
AIR 1608



ADITI KUMAR
(2021-25)
AIR 640



ALOK RANJAN
(2021-25)
AIR 1996



KOMAL KUMARI
(2022-26)
AIR 666



ANAND GUPTA
(2022-26)
AIR 2532



PAWAN KUMAR
(2021-25)
AIR 919



SANGITA KUMARI
(2021-25)
AIR 2756

IITs INTERNS



ANCHAL KUMARI
(2023-27)
IIT BHU



AYUSH ANAND
(2023-27)
IIT BHU



ANJANI GAURAV
(2023-27)
IIT BHU



GOURAV KUMAR
(2023-27)
IIT DHANBAD



ANSHU KUMAR
(2023-27)
IIT DHANBAD



HARSHIT RAI
(2023-27)
IIT BHU



ARYA SRIVASTAVA
(2023-27)
IIT DHANBAD



KHUSHI SRIVASTAVA
(2023-27)
IIT ROORKEE



**ASIF GULMOZZAM
FAYAZI**
(2023-27)
IIT BHU



NEETU DUTTA
(2023-27)
IIT BHU

IITs INTERNS



PALAK PRIYA
(2023-27)
IIT BHU



SANKALP SUMAN
(2023-27)
IIT DHANBAD



PAYAL JAISWAL
(2023-27)
IIT BHU



SANSKAR RAJ SINGH
(2023-27)
IIT BHU



PRIYA PANDEY
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IIT BHU



SHIVAM KUMAR
(2023-27)
IIT GUWAHATI



RAJ KUMAR BARNWAL
(2023-27)
IIT BHU



SHIVANGI KUMARI
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(2023-27)
IIT BHU



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3. Pratiksha Priya	2104058
4. Sangita Kumari	2104073
5. Vikas Kumar	2104091

Adani

1. Shilpa Kumari	2104111
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Dalmia Cement

1. Abhinandan Choudhary	2104006
2. Muskan Sahu	2104049
3. Neha Nandini	2104051
4. Siddhi Dubey	2104081

EcoLabs

1. Supriya Sharma	2104087
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Global Archer

1. Ajeet Kumar Mahato	2204001D
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Globus Spirit

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2. Balram Kumar	2104100
3. Kundan Kumar	2104043
4. Syed Shifran Ahmed	2104088

Indorama

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3. Ravi Kumar	2104109
4. Deo Dutta Kumar	2204004D
5. Rohit Kumar	2204009D
6. Vikash Tiwari	2204014D



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2104059

JSW

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2104069

Matix Fertilisers

1. Abhay Anand

2104003

2. Aman Ranjan Mishra

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3. Nisha Kumari

2104054

4. Raj

2204007D

Shyam Metals

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3. Satyam Pandey

2104075

4. Shuvam Banerjee

2104080

5. Suraj Kumar

2104113

6. Ujjwal Kumar

2104090



PLACEMENT REPORT

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2. Anshu Ranjan	2104020
3. Aryan Das	2104026
4. Ashish Kumar Saha	2104028
5. Indrani Das	2104037
6. Madhu Kumari	2104102
7. Puja Kumari Mahato	2104061
8. Suman Kumar	2104086
9. Unnati Singh	2104114
10. Vishwajeet Kumar	2104094

Tata Steel

1. Anupama Keshri	2104023
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TCS Ninja

1. Hritik Raj	2104036
2. Atanu Khawas	2204002D



PLACEMENT REPORT

Technip Energies

1. Aditi	2104007
2. Marzuqua Siddqui	2104105
3. Nikita Kumari	2104053
4. Rahul Chandra	2104062

TSUISL

1. Alok Ranjan	2104013
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VA Tech Wabag

1. Aditya Kumar	2104008
2. Manisha Kumari	2104045
3. Shalu Kumari	2104077

Vedanta

1. Aaron Rohi Lakra	2104002
2. Arnab Singh	2104024
3. Kashaf Fateema	2104038
4. Sneha Pandey	2104082
5. Vishnu Vaibhav	2104093



OUR RECRUITERS





OUR RECRUITERS



TATA STEEL UTILITIES AND INFRASTRUCTURE SERVICES LIMITED



RECENT EVENTS



QUIMICA'25

QUIMICA'25 is the annual flagship event of QUIMICA. This year, it featured two major competitions, Innoverse and Smart Scape along with three insightful webinars, creating a perfect blend of innovation, problem-solving, and knowledge-sharing.



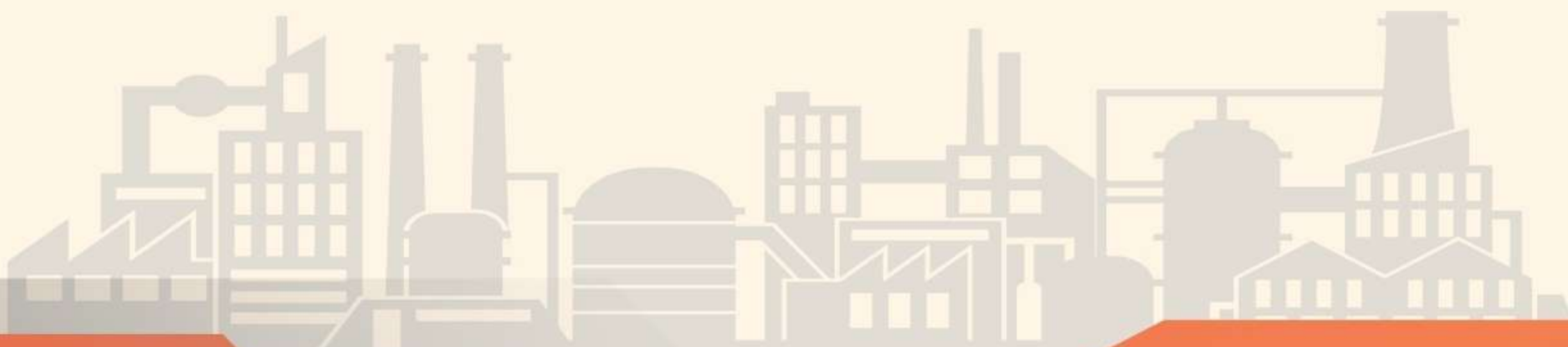


RECENT EVENTS



INNOVERSE

INNOVERSE: Innovation into Reality' is a thrilling three-round team event crafted to ignite creativity, test technical expertise, and highlight teamwork and presentation skills. Teams of 4-5 members tackle real-world industrial challenges, engage in dynamic debates, and deliver impactful presentations.



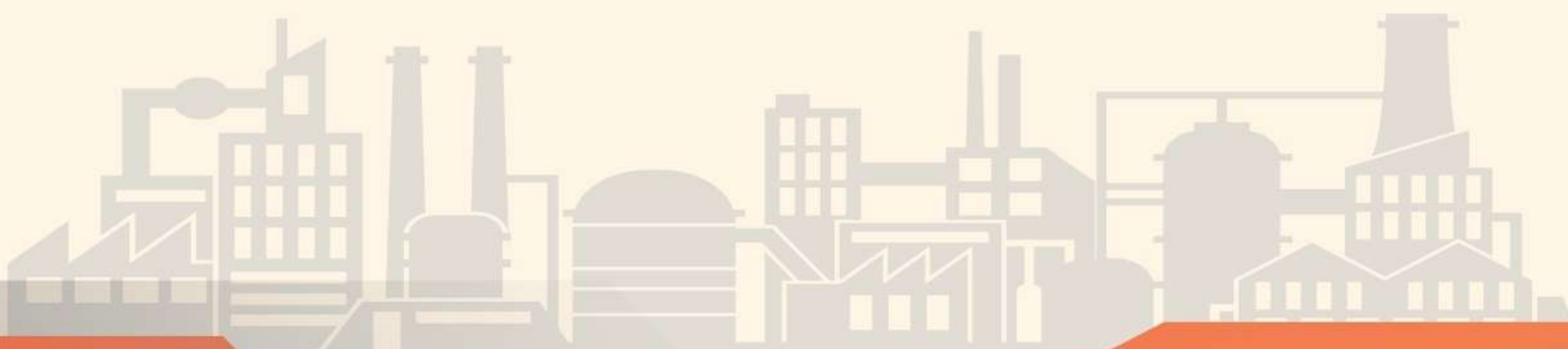


RECENT EVENTS



Smartscape

Smartscape: Smart scape stands out as one of BIT's most distinctive problem-solving competitions, designed to challenge analytical skills, teamwork, and quick thinking through puzzles, riddles, and layered problem statements. It transforms problem-solving into an adventurous journey, where innovation meets intellect and every challenge sparks a new way of thinking.





RECENT EVENTS

The poster for the BITS Blog event is presented by the Chemical Engineering Society. It features the BITS Pilani logo and the QUIMICA logo. The main title is "BITS BLOG" in a large, bold, blue font. Below it, the tagline "Bridging Campuses and Industries Through Words." is written in a smaller, black font. The event is open for entries from 17th Jan 2025. A blue button with a white arrow and the text "REGISTER NOW" is prominently displayed. A QR code is provided for scanning. The poster also includes contact information for Harshita Seth and Raj Kr. Barnwal, and social media links for the event on various platforms. The background of the poster shows industrial structures and a cityscape.

CHEMICAL ENGINEERING SOCIETY
Presents

BITS BLOG

"Bridging Campuses and Industries Through Words."

ENTRIES OPEN FROM
17TH JAN 2025

» REGISTER NOW

SCAN ME

CONTACT US:-
HARSHITA SETH: 7004199068
RAJ KR. BARNWAL: 7667010887

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BITS Blog

BITS Blog: The BITS Blog serves as an excellent platform for students to showcase their vision for the future of industries, where participants have presented thought-provoking perspectives that left a lasting impression. It continues to inspire innovation and encourages young minds to contribute fresh ideas for shaping tomorrow's world.





RECENT EVENTS



ElementaX

ElementaX is an engaging event that encourages scientific exploration and academic growth through three rounds: a Crossword Puzzle, Presentation, and Q&A. Promoting collaboration and knowledge sharing, it's perfect for passionate learners competing for the title of ElementaX Champion.





RECENT EVENTS



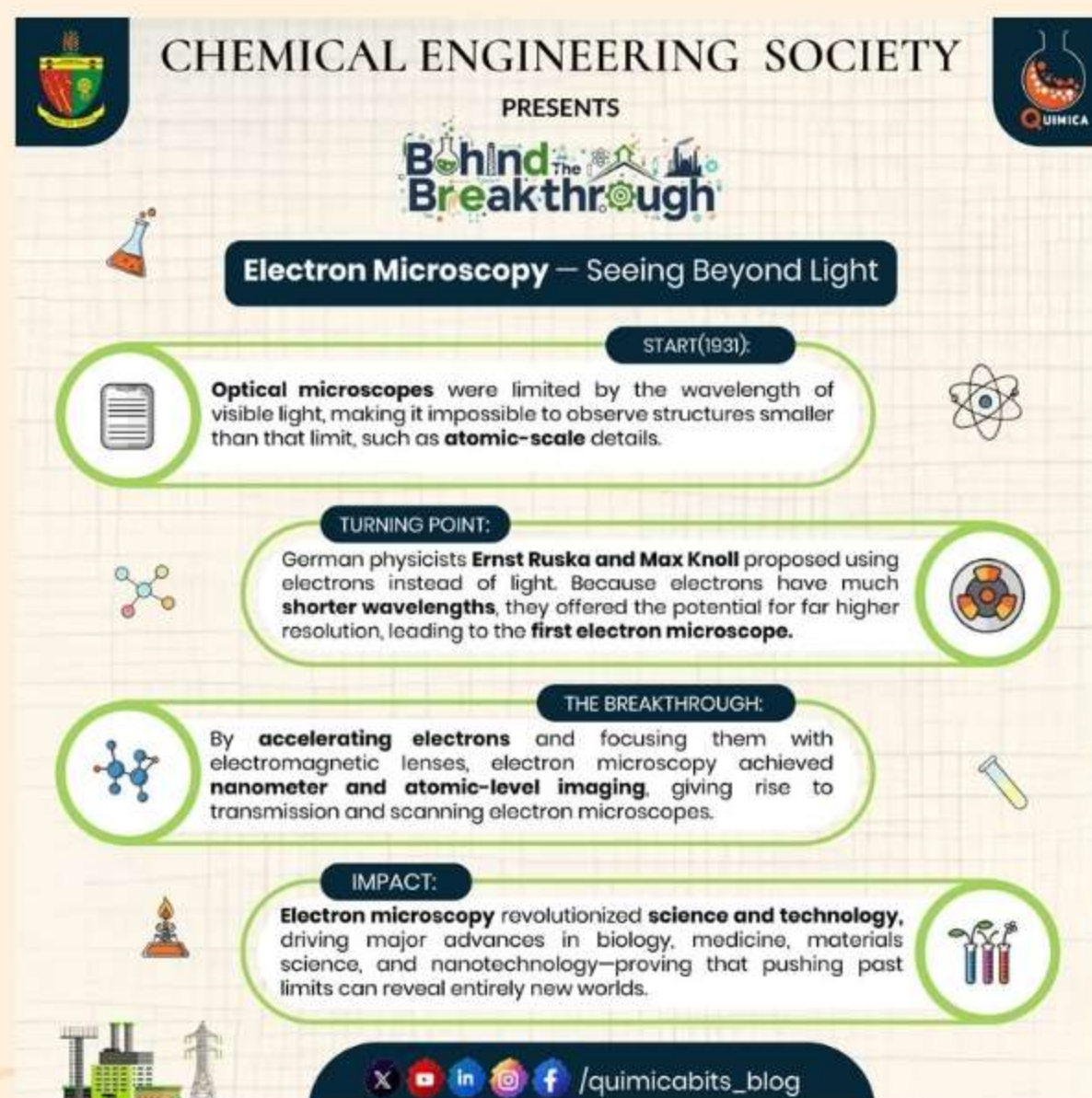
Jeopardy

The event “Jeopardy” is a dynamic and intellectually stimulating competition designed to test participants’ knowledge, creativity, and quick thinking. Featuring two engaging rounds—a pen-and-paper quiz and a Reverse Quiz Game, where participants create questions from given answers. By combining traditional quiz elements with an innovative twist, “Jeopardy” offers a unique and enriching experience.





NEW SEGMENTS

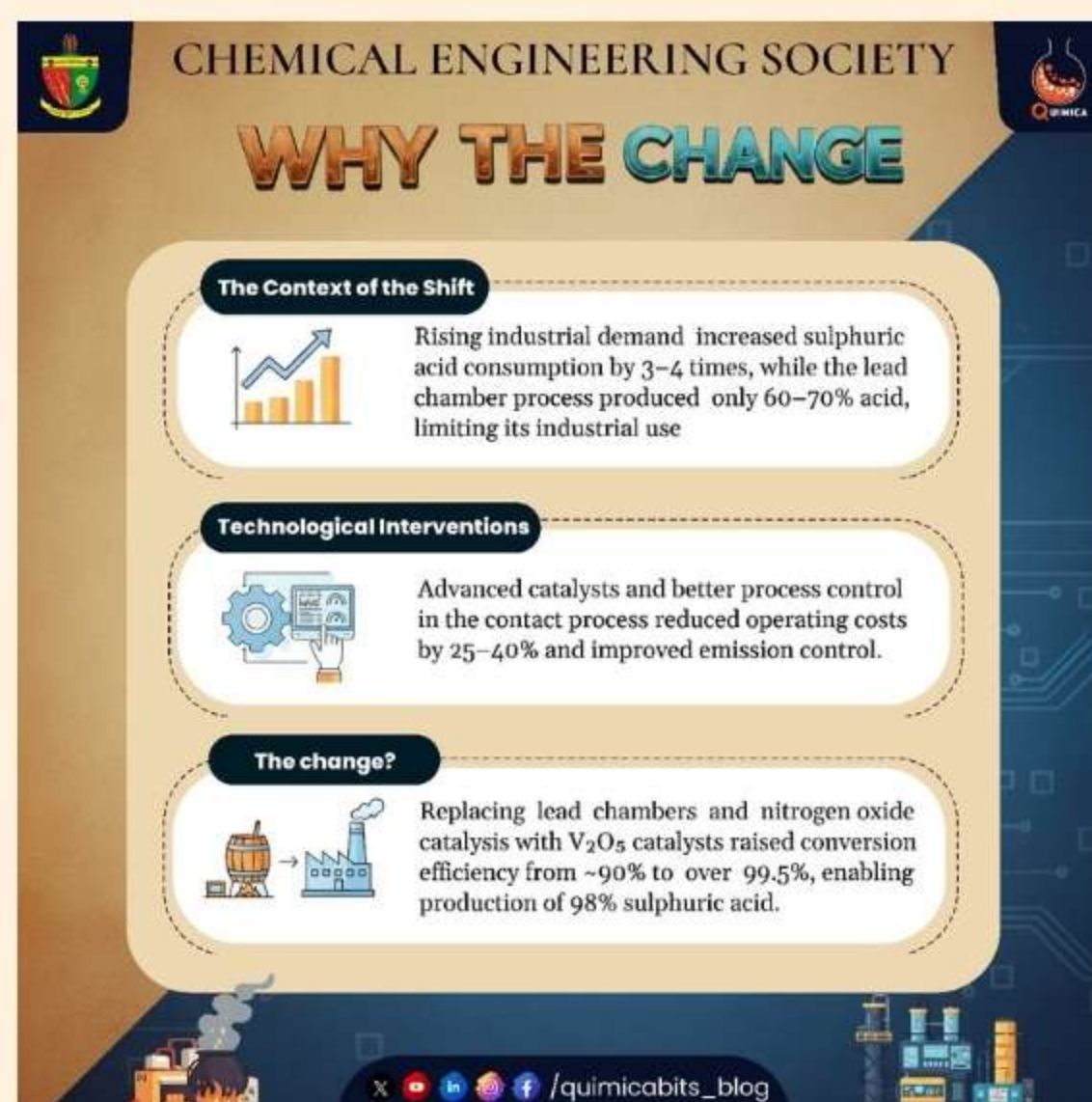
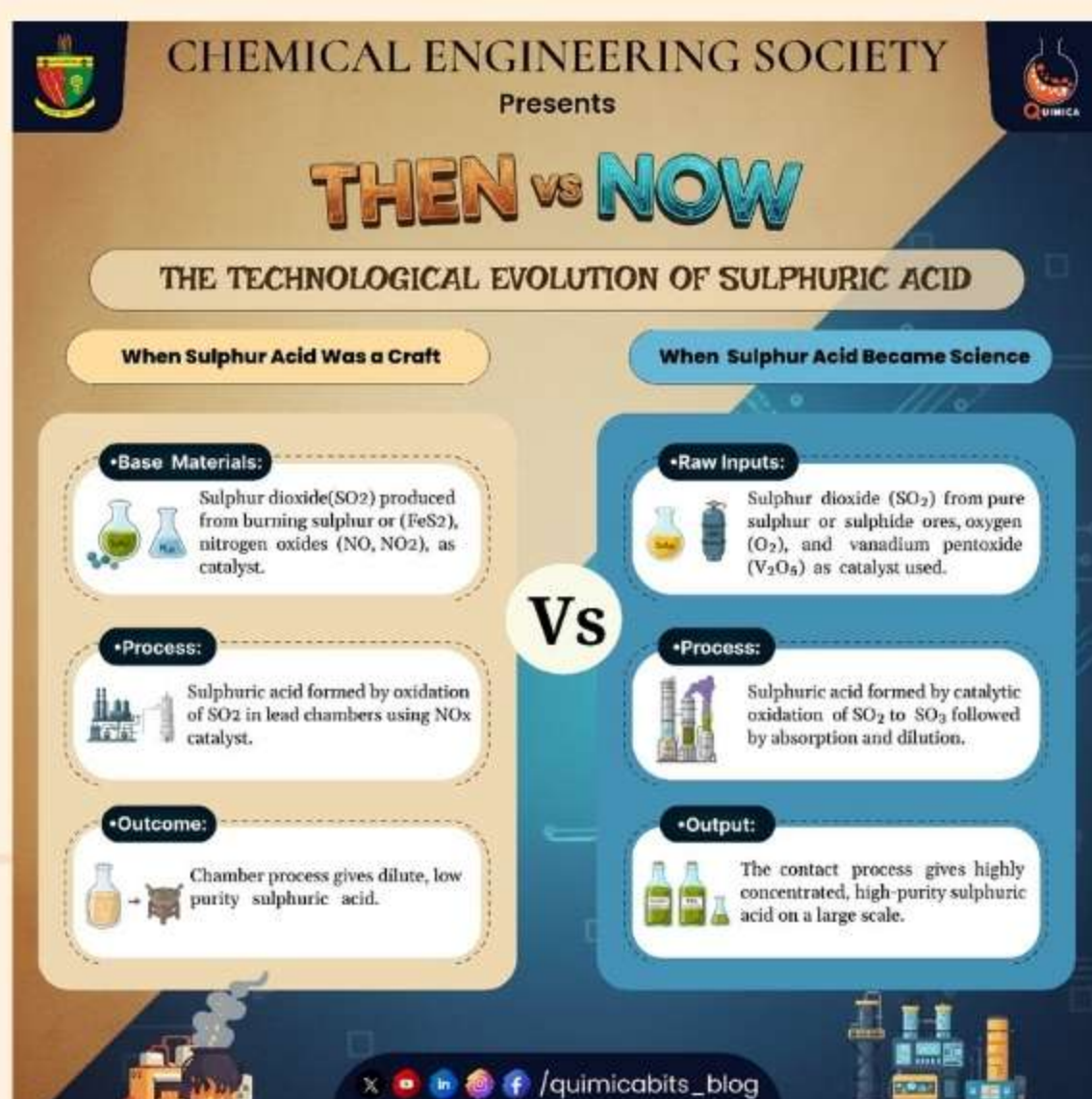


BEHIND THE BREAKTHROUGH

Behind the Breakthrough dives into the stories behind major discoveries and innovations in Chemical Engineering. It highlights the challenges, ideas, and experiments that led to technological progress. The segment helps students appreciate the journey of innovation, not just the final outcome. It connects fundamental concepts with real-world industrial impact.



NEW SEGMENTS



THEN VS NOW

Then vs Now presents a comparative insight into past and present practices in Chemical Engineering. It showcases the evolution of processes, equipment, and efficiency over time. The segment helps readers understand the role of technological advancement in industry growth. It effectively links historical developments with current engineering practices.

NEW SEGMENTS



CHEMSHOT

Chem Shot is our exclusive video segment that dives deep into the dynamic world of Chemical Engineering. We explore industry processes, cutting-edge equipment, and emerging trends, offering viewers a comprehensive understanding of how Chemical Engineering shapes industries and the world around us.



NEW SEGMENTS

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CHESS: Chemical Engineering Sunday Series.

"Move your knowledge forward, one Sunday at a time." ...more

Darcy–Weisbach equation is used to calculate:
The author can see how you vote. [Learn more](#)

A. Flow rate	32%
B. Pressure drop	56%
C. Reynolds number	6%
D. Velocity	6%

34 votes • Poll closed

  26 1 comment

CHESS

QUIMICA also conducts CHESS (Chemical Engineering Sunday Series), a weekly interactive segment featuring thought-provoking questions every Sunday. The initiative engaged students beyond classrooms, fostering curiosity, active participation, and healthy competition.

> ALUMNI'S ARTICLE



Mr. Nitish Kumar
Chemical Engineering,
2019

A biogas plant is to be set up near Hostel 26 to reuse the food waste from Hostel 7, Hostel 8, and Hostel 26. The biogas produced will be utilized in Hostel 26 mess in replacement of LPG. Mess will be charged as per the current market price of LPG/kg. The running of the plant and monthly collection of biogas bills from mess owners will be done by a professor in charge. Electricity bills, water bills, and plant maintenance will be done using the profit made by monthly collections. Urja Viable Solution (UVS) will promote the sustainable production of renewable energy from the biogas obtained from food and beverage industry waste (agro-food waste) in small-scale concepts for energy self sufficiency. This action will contribute to secure, sustainable, and competitively priced energy by promoting new and renewable energy sources and supporting energy diversification. The project strategy is designed for the target group, the agro-food sector, which is at the same time the energy producer and the demand-side with the proposed scheme:

> ALUMNI'S ARTICLE

First, the needs of end-users in terms of energy demand will be analyzed, as well as the difficulties they encounter when considering to install a biogas production facility.

Second, the necessary tools to address these needs will be developed according to the gathered information, including business collaboration models, small-scale process design and promotion, energy demand management models, and training to build-up of skills, awareness and networking.

Third, on-field actions will be implemented to promote this small-scale concept and to bring the developed tools to the end-users, including training sessions (face-to-face and on-line), workshops, webinars website, etc.

Specific programme of face-to-face activities with the most promising implementation sites will be carried out. UVS will produce a significant impact in terms of renewable energies uptake.

It will act as catalyst and replication of success, triggering decisions to invest in renewable energy production from agro-food industry waste. Total transferability is guaranteed between participant countries that cover the most relevant regions of Hostels in terms of agro-food waste generation. The high relevance of the participants and their dissemination capacity will assure the high visibility of the BIOGAS - Sustainable small-scale biogas from agro-food waste for energy self-sufficiency 2 actions, making intelligent energy technologies as widely available as possible. A train-the-trainer approach ensures impacts beyond the actual project duration.

> ALUMNI'S ARTICLE



Mr. Keshu Ranjan
Chemical Engineering,
2018

Quimica has provided me with a wonderful environment and opportunity to learn and grow myself academically as well as to secure a place in the corporate world. This is the time for all of you to understand the importance of any society with a vision of interacting more and more Chemical Engineers across the country. This college and my department has always guided and supported me for reaching my goal. The professors and seniors left no stone unturned to develop my communication and soft skills. Today, students from our department and our college are present almost in every corner of the world from various reputed institutions and research organisations to big shot corporate setups. Such efforts have inspired our current lot of students. These clubs and societies enhance the student's interactive skills on a professional level and help in personality development.

I always cherish the memories and the favourite pastimes I had while working as a part of the Quimica family. I feel immense felicity to have worked together as a team to convert every single opportunity. I always look forward to keeping in touch with all my fellow members. B.I.T. Sindri and my department will always be an inseparable part of my life for sure.

Recent Advancements in Chemical Industry

Blockchain: A revolutionary development in the chemical sector.
By: Saurabh Mishra, 2019 Batch.
Co-Convenor, Quimica.

Blockchain is a decentralized peer-to-peer digital networking technology that can support the chemical sector throughout its entirety, from establishing a cohesive trading framework to the creation of client-focused products.

Trust without Reliance

"2018 Technology Vision," 77% of executives in the chemical industry said that integrating blockchain into firm systems would take one to three years, and 71% agreed that blockchain will have a significant impact on their business during the following three years. It transfers the trust factor from the participant to the system itself and is sometimes referred to as the "trustless trust" configuration. The necessity of integrating blockchain technology into business processes is increased by the integral-cum-static design. Pioneers from a variety of industries, including O&G, Chemical, and Pharma, have begun deploying this technology at various levels to reap industry-specific benefits, such as streamlining workflow processes, tokenizing the value of their final products' recycling, and improving supply chain legality.



Recent Advancements in Chemical Industry

Water Purification using Nanotechnology

By: Akash Deep, 2019 Batch.

Only 0.5 percent of freshwater is transportable. Filtration of the contaminated water is deemed necessary for human consumption. In our daily lives, we come across a variety of technologies, such as UV filtration and reverse osmosis (RO). Nanotechnology has been developed by scientists to make water portable and free of contaminants and undesirable minerals (RO). The process of working with atoms at the nanoscale is known as nanotechnology.

The efficiency :

Compared to traditional methods like boiling, filtration, distillation, sedimentation, chlorination, and oxidation, nanotechnology is more efficient. This process softens water by eliminating chemical, biological, and physical contaminants using two membranes. Additionally, this very successful technique has been commercialized for safe drinking water, such as Nanoceram, NanoH₂O, etc.

Our features :

(a) Less volume and more surface area.

(b) At the nanoscale, membrane material can alternatively its electrical, physical, chemical, optical, or biological characteristics.

(c) Reactions in chemistry and biology are simpler. Nanocellulose-based filtration is renewable and has great potential in the field of nanotechnology for water purification. There are various materials available, including cellulose nanofibrils (CNF) and cellulose nanocrystals (CNC). Sulfuric acid hydrolysis and mechanical grinding are the methods used to prepare the nanocellulose material. Absorption is the foundation of the standard water purification system. Even though the nanotech approach is very effective, it is expensive and has limitations in commercial production.

Research on the use of nanotechnology in water purification is crucial. Fresh water is becoming increasingly important for future water purification systems due to its declining quality for human consumption.



Recent Advancements in Chemical Industry

Quantum Computing

By: Anjali Sah, 2019 Batch.

The chemical sector is ready to begin reaping the benefits of quantum computing's significantly increased modeling and computational power. Quantum computing has received a lot of attention in recent years.

There have been numerous reports of work being done in this area at tech behemoths like Google and IBM, and players from a variety of industries, including the chemical industry, have made investments in the field of quantum computing.

Adoption :

This sector has adopted digital innovations and practices that are permeating business and society relatively late. By utilizing the principles of quantum physics, quantum computing is able to perform some calculations far more quickly than traditional computers.

NOBEL PRIZE IN CHEMISTRY



SUSUMU KITAGAWA



RICHARD ROBSON



OMAR M.YAGHI

The 2025 Nobel Prize in Chemistry was awarded to Susumu Kitagawa, Richard Robson, and Omar M. Yaghi for their pioneering work on metal-organic frameworks (MOFs). MOFs are highly porous crystalline materials formed by linking metal ions with organic molecules, resulting in exceptionally large internal surface areas.

These materials have opened new possibilities in gas storage and separation, carbon capture, water harvesting from air, catalysis, and clean energy technologies. Their ability to be precisely designed for specific applications makes them highly valuable in addressing environmental and energy-related challenges. The recognition of MOFs highlights their transformative impact on modern chemistry and their significant role in building a more sustainable future.

NOBEL PRIZE IN CHEMISTRY



DAVID BAKER

The 2024 Nobel Prize in Chemistry went to David Baker, a renowned professor at the University of Washington, for his innovative work in computational protein design. Using computational methods he invented, Baker successfully created a new protein in 2003 that was unlike any discovered in nature. This accomplishment paved the way for a vast array of novel proteins, each suited to a particular purpose.

Baker's work has significant ramifications for many different industries. His designed proteins have improved disease treatment and prevention in the pharmaceutical industry by producing more potent medications and vaccines. These proteins help create advanced materials with special qualities in anomaterials, which spurs technological and manufacturing innovation.

Baker's accomplishments highlight the revolutionary potential of computational methods in biochemistry, ushering in a new era of protein engineering with significant societal advantages.

NOBEL PRIZE IN CHEMISTRY



DEMIS HASSABIS



LOUIS E. BRUS



ALEKSEY YEKIMOV

In 2023, the Nobel Prize in Chemistry was awarded to Moungi G. Bawendi, Louis E. Brus, and Aleksey Yekimov for their groundbreaking work on the discovery and controlled synthesis of quantum dots. Quantum dots are ultra-small semiconductor nanoparticles whose optical and electronic properties are governed by their nanometer-scale dimensions.

These materials have brought a major transformation in science and technology. One of their most significant features is the ability to emit highly bright and size-tunable colors when excited by light. This property has enabled their widespread use in modern display technologies, particularly QLED televisions, delivering enhanced color accuracy, brightness, and energy efficiency.

The contributions of Bawendi, Brus, and Yekimov highlight the immense potential of nanotechnology. By precisely manipulating matter at the atomic and molecular levels, their work has paved the way for innovations across diverse fields, including electronics, biomedical imaging, photovoltaics, and quantum computing, marking a significant advancement in modern chemical science.

NOBEL PRIZE IN CHEMISTRY



DEMIS HASSABIS



JOHN JUMPER

Demis Hassabis and John Jumper, prominent researchers at Google DeepMind, were awarded the 2024 Nobel Prize in Chemistry for their pioneering application of artificial intelligence in protein structure prediction. They developed AlphaFold, a revolutionary AI-based system capable of predicting the three-dimensional structures of proteins with exceptional accuracy.

This breakthrough addressed one of the most complex challenges in molecular science—understanding how proteins fold into their functional forms. Accurate knowledge of protein structures is essential for gaining insights into disease mechanisms, accelerating drug discovery, and advancing biotechnological innovations.

The Nobel Committee recognized their work as a landmark achievement bridging chemistry, biology, and artificial intelligence. Their contribution highlights the immense potential of AI-driven approaches in solving complex scientific problems and signifies the beginning of a new era of interdisciplinary research and innovation.

GALLERY



GALLERY



GALLERY





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WITH OUR DEEPEST GRATITUDE...

WE'RE EXCITED TO INTRODUCE, QUIMICABITS, OUR OFFICIAL NEWSLETTER, WHICH WOULD HELP YOU ALL TO FOSTER A FEELING OF COMMUNITY AND ATTACHMENT TOWARDS OUR CHEMICAL DEPARTMENT.

WE OWE A HUGE DEBT OF GRATITUDE TO EVERYONE WHO PUT IN COUNTLESS HOURS TO CREATE THIS MAGAZINE AND MAKE IT A PERFECT WORK OF ART.

WE WOULD LIKE TO THANK OUR DIRECTOR PROF (DR.) D.K.SINGH SIR AND HOD CHEMICAL ENGINEERING - PROF (DR.) AMIT KUMAR GUPTA WHOSE EXCEPTIONAL TECHNICAL SUPPORT AND LOGISTICAL ASSISTANCE ACTED AS A BEACON OF GUIDANCE FOR US.

THE GREATNESS OF OUR INSTITUTE AND PUBLICATION HAS BEEN EXCLUSIVELY ATTRIBUTED TO OUR DISTINGUISHED FACULTY, WHO HAVE MADE IT POSSIBLE FOR US TO PUBLISH THIS MODEL OF EXCELLENCE. WE WOULD LIKE TO SHOW OUR SINCERE GRATITUDE TOWARDS THEM.

ALSO, WE WOULD LIKE TO EXPRESS OUR GRATITUDE TO THE SHREWD AND INVENTIVE ALUMNI, WHO GAVE US HOPE THROUGHOUT THIS ROUGH ROAD, WHICH WOULD HAVE PREVENTED US FROM GETTING THIS FAR. WE SINCERELY THANK EVERY ONE OF THEM FOR THEIR EXTRAORDINARY WORK AND COMMITMENT.

ADDITIONALLY, WE WOULD LIKE TO SHOW OUR THANKS TO THE POST- BEARERS OF THE CHEMICAL ENGINEERING SOCIETY FOR INSPIRING US TO GO ABOVE OUR BOUNDARIES AND FOR HELPING US TO DEVELOP THIS PIECE OF EXCELLENCE. WE ARE LUCKY TO HAVE STUDENT COORDINATORS WHO ARE SO BRILLIANT AND COMPETITIVE AND WHOSE EXCEPTIONAL ACCOMPLISHMENTS ARE AN INSPIRATION TO US.

AFTER GOING THROUGH THIS MAGAZINE, YOU WILL COMPLETELY COMPREHEND THE POTENTIAL OF THE CHEMICAL ENGINEERING DEPARTMENT, WHICH YOU WILL ALL ADMIRE IN THE FOLLOWING YEARS.